

Hsin Ya Lin

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WORK EXPERIENCE

Satellite Electronics Intern | Taiwan Space Agency | Jul 2026 to Aug 2026

- Verified the RS422 link byte by byte by decoding captured wire traffic against an independent implementation, establishing 7 protocol facts that had been assumed and diagnosing 4 firmware defects including memory corruption behind a reboot loop.
- Wrote roughly 9,000 lines of real time C and C++ firmware for that board, and cut combined communication and motor latency from 727ms to 263ms.
- Designed the satellite reaction wheel test board in KiCad and built it, integrating an ESP32-S3 microcontroller, a 9 axis IMU over I2C, an RS422 transceiver and a DC-DC buck converter with hardware and software safety interlocks.

RESEARCH

Digital Design and Verification | Purdue SoCET RISC-V SoC Team | Aug 2026 to Present

- Designing the SPI controller in SystemVerilog that reads an inertial sensor into a RISC-V system on chip, on a team building its own on-chip data bus with memory mapped IO after evaluating and rejecting an existing one.
- Moving quaternion state estimation out of software and into a hardware Kalman filter accelerator on the same die, checked against a golden model and profiled on cycle count, power and area against a software-only baseline on the same RISC-V core.

Digital Design and Verification | Purdue SoCET Matrix Coprocessor Team | Jan 2026 to May 2026

- Designed a matrix math coprocessor in SystemVerilog running on FPGA hardware, with a 33 state main control FSM, nested FSMs inside the arithmetic modules, and a memory map split three ways so operands and results cannot overwrite one another, verified with 7 module level testbenches.
- Carried the same RTL through an ASIC flow covering logic synthesis in Yosys, place and route, timing closure and standard cell layout in Cadence Virtuoso, with simulation in Icarus Verilog and waveform analysis in QuestaSim.

TECHNICAL EXPERIENCE

Power and Hardware Engineer | Purdue Lunabotics | Sep 2025 to Aug 2026

- Designed a 24V, 20A six cell battery sensing and protection board around the TI BQ77915 in KiCad, sizing the external resistor network that sets the overcurrent and overvoltage thresholds, and simulated the MOSFET switching and RC values in LTspice with a simplified model of the protector.
- Laid the board out for fabrication, hand assembled and micro soldered 50+ components onto it, diagnosed voltage and thermal variances during bench test, and integrated it into the rover power distribution system.

EDUCATION

Purdue University | West Lafayette, Indiana | Aug 2025 to Present

Bachelor of Science in Electrical Engineering (expected December 2028)

GPA: 3.98 / 4.0; Dean's List (2 semesters) & Semester Honors (2 semesters)

SKILLS

- Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB, Assembly, shell scripting, Linux/Unix, Git
- Silicon: RTL design and verification, FSM and datapath, module level testbenches, functional simulation, waveform analysis, signal tracing, golden model comparison, logic synthesis, place and route, timing closure, standard cell layout, FPGA prototyping, RTL to GDS
- SoC: RISC-V, hardware accelerator design, on-chip bus design, memory mapped IO, memory map partitioning, cycle count, power and area profiling against a software baseline
- Embedded: bare metal C and C++ real time firmware, microcontroller bring-up, ESP32-S3 and MSP430, peripheral and sensor drivers, safety interlocks, SPI, I2C, UART, RS422, CAN bus, byte level protocol verification against captured wire traffic
- Hardware and bench: schematic through PCB layout and assembly in KiCad and Altium, micro soldering, surface mount rework, DC-DC buck and boost conversion, battery protection, analog filters and amplifiers, hardware in the loop testing, parameter identification from captured data, oscilloscope, logic analyzer, spectrum analyzer, LTspice
- Tools: Cadence Virtuoso, Yosys, QuestaSim, ModelSim, SimVision, Icarus Verilog, GTKWave, gdb, MATLAB and Simulink, MATLAB Coder, PyTorch, KiCad, Altium, LTspice